

Feature Selection

- **Filter methods:** Features are selected on the basis of their scores in various statistical tests for their correlation and with the outcome variable e.g., Chi squared test (for categorical variables) or Pearson correlation coefficient scores (for continuous variable).
- **Wrapper methods:** Try to use a subset of features and train a model using them. Based on the inferences from the previous model, decide to add or remove features from the subset. The problem is essentially reduced to a search problem. Wrapper methods are usually computationally expensive due to training multiple times e.g., recursive feature elimination algorithm.

Feature Selection

- **Recursive Feature elimination:** Greedy optimization algorithm to find the best performing feature subset. It repeatedly creates models and keeps aside the best or worst performing feature at each iteration. It constructs the next model with the left features until all the features are exhausted. It then ranks the features based on the order of their elimination.
- **Embedded methods:** Learn which features best contribute to the accuracy of the model while the model is being created. The most common type of embedded feature selection methods are regularization methods.

Dimensionality Reduction

- Transform high dimensional data into lower dimensional space
- Transformation will lead to information loss
- Goal: Keep the information loss to minimum

Data Preprocessing

- Data mining a buzzword is predecessor to Data science
- A popular methodology in data mining is Cross Industry Standard Process for data mining (CRISP DM)
- CRISP DM has the following phases:
 - Business understanding: Business person formulates a business problem (domain experts)
 - Data understanding: A technical analyst also gets involved along with domain experts. Usually called exploration phase involves understanding complicated data

Data Preprocessing

- Data Preparation: Phase where we create training and test datasets. Can call Preprocessing phase
- Modeling: Formulate a model to fit data and is usually associated with Machine Learning
- Evaluation: Evaluate the model and data to check if the business problem is solved
- Deployment: Setting up the system in a production environment
- Key to all this is high quality data
 - Cleaning data is therefore important
 - Cleaning data can form large part of ML many times

Data Preprocessing

- No one way to clean data, involves manual work
- Visualize the data
- Data can be grid of numbers, images, audio, text or something else
- Grid of numbers is most convenient form and hence ideal to work to have numerical features
 - Table of numbers is assumed default unless specified otherwise

Feature Engineering

- Features can be of many types
- Feature values can follow distributions e.g., normal, binomial, poisson etc.
- Features can be binary yes/no, +/- etc.
- Features can be categorical e.g. instance of continents, high/medium/low
- Features can be quantitative e.g. temperature in degrees or price in dollars
- Feature engineering is process of creating or improving features

Feature Engineering

- Features often based on common sense, domain knowledge or prior experience
 - More art than science
- Missing values: Data can have missing values for some features
 - Sometimes easy to find by just scanning, counting etc
- How to deal with missing values ?
 - Ignore missing values [works sometimes]
 - Substitute missing values by fixed values called imputing e.g. impute the arithmetic mean, median or mode

Feature Engineering

- Humans deal with various types of values, ML algorithms in general deal with numerical values
- **Label Encoding:** Refers to converting labels into numeric form to make it ML readable
 - E.g., Height can have Tall, Medium and Short as categories
 - Using label encoding can convert to 0, 1 and 2
- Process can be called **integer-label encoding**
 - Issue: Learner may conclude there is an ordering

Feature Engineering

- **One-hot-encoding or one-of-K:** Refers to splitting the column which contains numerical *categorical data* to many columns depending on the number of categories present in that column.
 - Each column contains “0” or “1” corresponding to which column it has been placed.

Feature Engineering

Fruit	Categorical value of fruit	Price
apple	1	5
mango	2	10
apple	1	15
orange	3	20

- After one hot encoding

apple	mango	orange	price
1	0	0	5
0	1	0	10
1	0	0	15
0	0	1	20

Feature Engineering

- **Scaling:** Values of features can differ by orders of magnitude. May need to scale many times. Some common strategies:
 - **Standardization:** Remove mean of feature and divide by standard deviation – if feature values normally distributed, we get Gaussian centered around 0 with variance of 1
 - If feature values not normally distributed, remove median and divide by the interquartile range i.e. range between the first (25th) and third (75th) quartile
 - **Scaling** features to range is commonly used – common range used is 0 to 1

Feature Engineering

- **Polynomial features:** Given two features a and b , polynomial relation of form $a^2 + ab + b^2$: ab is called interaction
 - Per Occam's razor, we want to avoid higher order polynomials and interaction of many features
 - Complex poly's can be hard to compute for little additional benefit

Feature Engineering

Location	Size	Prediction
good	big	300,000
good	small	200,000
bad	big	250,000
bad	small	150,000

- Can decompose the model prediction as:
 - Constant term (150,000),
 - An effect for the size feature (+100,000 if big; +0 if small)
 - An effect for the location (+50,000 if good; +0 if bad).
 - This decomposition fully explains the model predictions.

Feature Engineering

Location	Size	Prediction
good	big	400,000
good	small	200,000
bad	big	250,000
bad	small	150,000

- Can decompose the prediction table as:
 - Constant term (150,000)
 - An effect for the size feature (+100,000 if big, +0 if small)
 - An effect for the location (+50,000 if good, +0 if bad).
 - Need an additional term for interaction: +100,000 if the house is big and in a good location.

Feature Engineering

- **Power transformations:** Functions to transform numerical features into a more convenient form e.g. conform better or closer to normal distribution.
 - Box-Cox transform tries to find best power needed to transform into data that is closer to normal distribution
- **Binning:** Separate feature values into bins e.g. precipitation values can be converted into rain or no-rain or high/medium/low/no bins.
 - Binning can lead to loss of information but improve on speed and memory requirements
 - Can reduce chance of overfitting